UNIT FIZIK
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Semester 2 Sesi 2023/2024

*Answer **all** questions.*

1 (a)

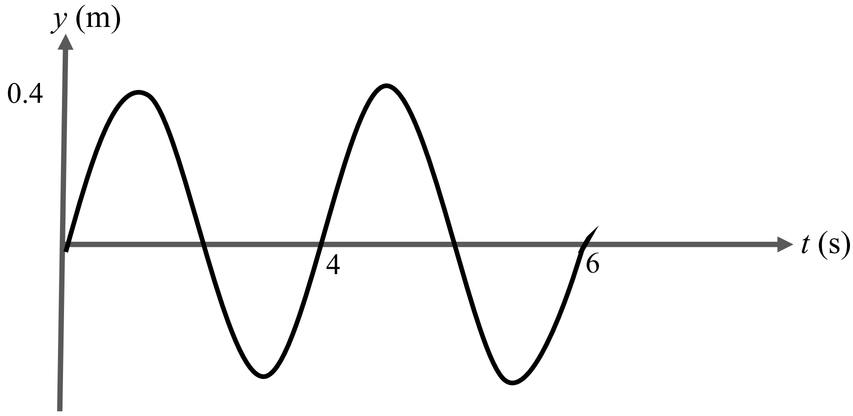


FIGURE 1

FIGURE 1 shows a graph of displacement, y versus time, t of a progressive wave propagates to the positive x-axis. If wavelength given is 0.4m, determine the

(i) amplitude.
(ii) frequency.
(iii) wave number.
(iv) progressive wave equation.
(v) wave propagation velocity.
(vi) particle vibrational velocity at $x = 1.0$ m and $t = 2.0$ s.

[12 *marks*]

(b) Two sinusoidal waves traveling in opposite directions interfere to produce a standing wave with the wave function

$$y\,(x,t) = 2 \sin 1.5x \cos 250t$$

where y and x are in metre and t is in second, determine the

(i) wavelength
(ii) frequency
(iii) speed of waves

[4 *marks*]

2 (a) An electric oven indicates a current flow of 12.0 A when connected to 240V supply.

(i) What is the resistance of the oven?
(ii) How much charge passes through it in 5 minutes?

[3 *marks*]

(b) A piece of aluminium rod has 2.0 cm diameter and 10.0 cm long.
(i) What is the resistance of the rod at room temperature?
(Given the resistivity of aluminium rod, $\rho_{\text{aluminium}} = 2.8\ x\ 10^{-8}$ Ωm)
(ii) Determine the current flows through the rod when a potential difference 24 V is applied to it.

[3 *marks*]

(c)

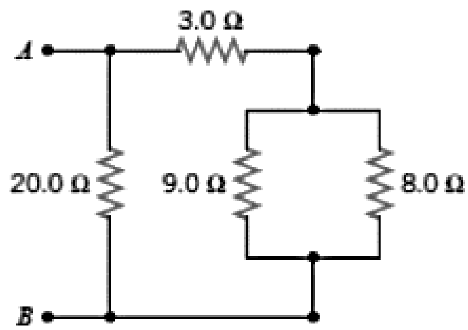


FIGURE 2

From the **FIGURE 2** above, determine the

(i) equivalent resistance between the point A and B
(ii) current flow through 3 Ω resistor if V_{AB} =10 V

[6 *marks*]

3 (a) Two conductors having net charges of +20.0 μC and –20.0 μC have a potential difference of 10.0 V between them. Determine the capacitance of the system.

[2 *marks*]

(b)

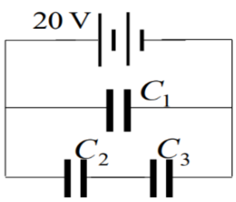


FIGURE 3

FIGURE 3 shows a combination of three capacitors where C_1 = 200 μF, C_2 = 220 μF and C_3 = 57 μF. A 20 V supply is connected to the combination. Determine

(i) the effective capacitance in the circuit.
(ii) the charge stored in the capacitor C_1.

[5 *marks*]

(c) A $24\,\mu F$ capacitor is charged to a potential difference of $6\,V$. It is then disconnected and discharges through $10\,k\Omega$ resistor. Calculate

(i) the time constant

(ii) the charge remained in the capacitor after $0.6\,s$.

[3 *marks*]

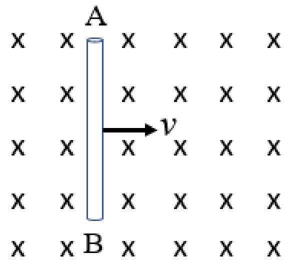
4\. (a) A current of 6.2 A flows in a long straight wire in a free space. Determine the magnitude of the magnetic field at a point which is at a perpendicular distance of 3.4 cm from the conductor.

[2 *marks*]

(b) A circular coil is carrying a current of 25 mA. If the magnitude of magnetic field at the centre of circular coil is 1.0×10^{-6} T, calculate the radius of the circular coil.

[2 *marks*]

(c)



FUGURE 4

FIGURE 4 shows a 1.5 m conducting rod AB is placed in a uniform magnetic field with magnitude of 4.2 T. The rod is pulled to the right with a uniform velocity of 3.5 m s^{-1}. Determine the

(i) magnitude of induced emf in the rod

(ii) direction of induce current

[3 *marks*]

(d) A circular coil of 100 turns and cross-sectional area of 0.15 m^2 is placed in a uniform magnetic field of 30 mT. Determine the

(i) magnetic flux linkage through the coil if the area of the coil parallel to the magnetic field.

(ii) magnitude of induced emf if the cross-sectional area of the coil change to 0.08 m^2 in 2 s.

[7 *marks*]

5\. (a) An upright image is produced by a spherical mirror at a distance of 18 cm behind the mirror. If object is placed 4.5 cm in front of the mirror, determine the magnification **and** type of spherical mirror used.

[2 *marks*]

(b) A spherical mirror is located 12 cm from an object produced an upright image one-third as large as that object. Determine the
(i) image distance
(ii) focal length
(iii) radius of curvature of the mirror
(iv) type of mirror

[5 *marks*]

(c) A convex lens has a focal length of 5.0 cm. If the image is real and the image distance is one third of the object distance, calculate the

(i) object distance.
(ii) magnification of the image.

[4 *marks*]

(d) A virtual image with twice the diameter is formed when a ruler is positioned 3.0 cm in front of a convex lens. Calculate the focal length of the lens.

[3 *marks*]

6 (a) In a Young's double slit experiment, a yellow monochromatic light of wavelength 680 nm shines on the double slit. The separation between the slits is 0.55 mm and it is placed 2.50 m from a screen. Calculate the

(i) separation between the zeroth-order maxima and first-order maxima.
(ii) separation between the second-order maxima and fourth-order maxima.

[6 *marks*]

(b) Two slits with separation of 0.30 mm are illuminated by light 580 nm and the interference pattern is observed on screen 3.00 m from the slits. Calculate the

(i) distance of second dark fringe from central bright.
(ii) distance between the second dark fringe and second bright fringe.

[8 *marks*]

END OF QUESTIONS